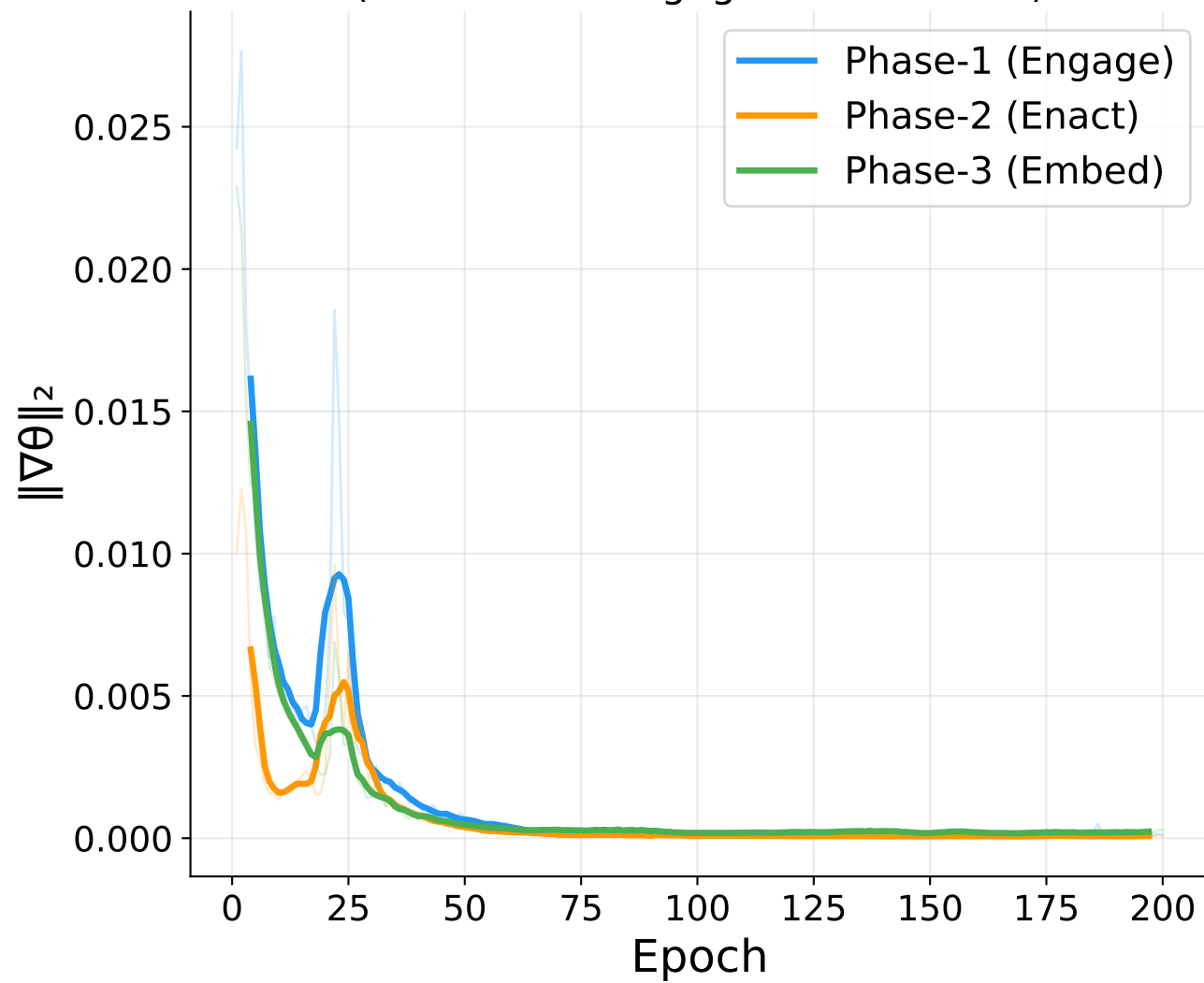
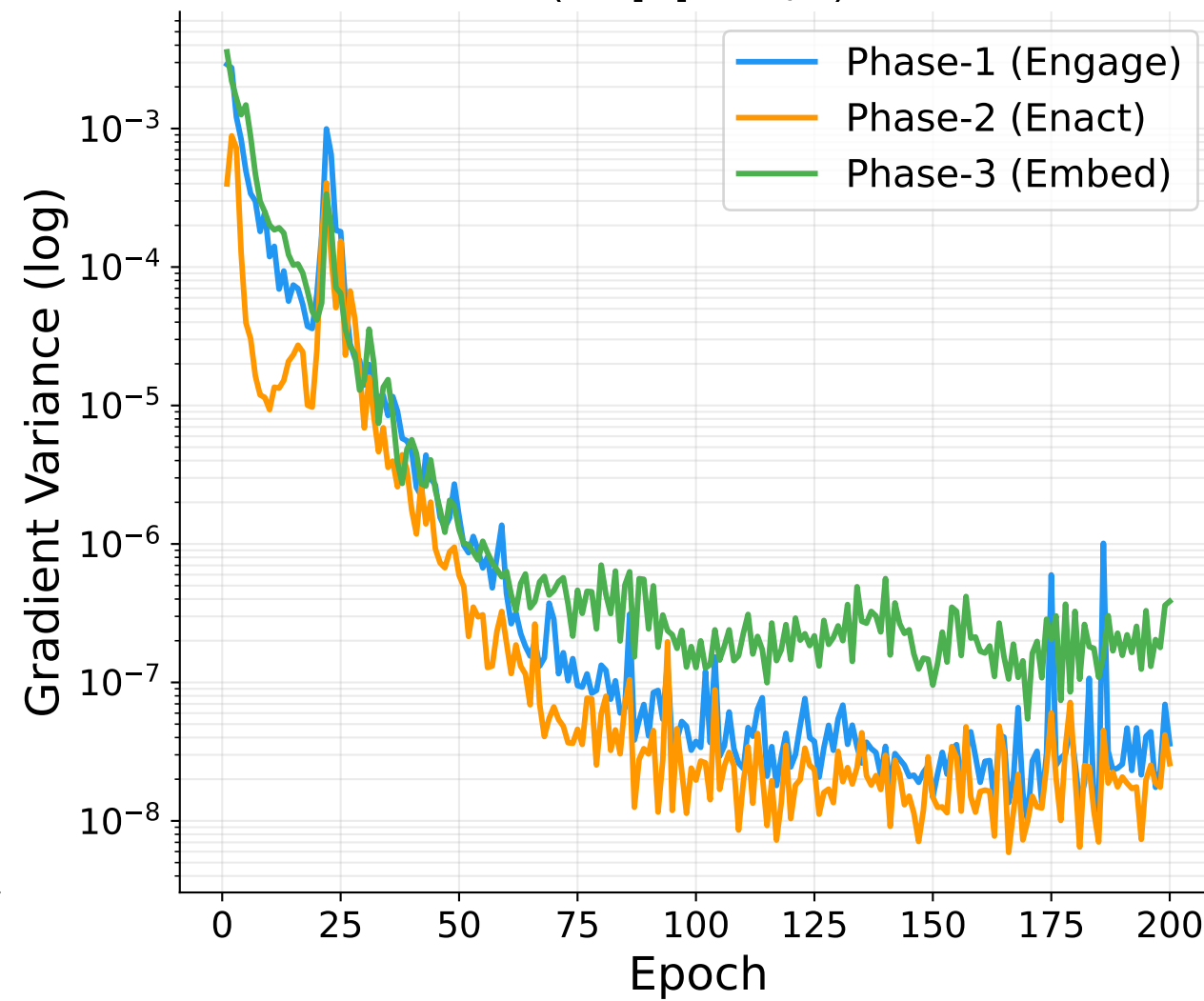


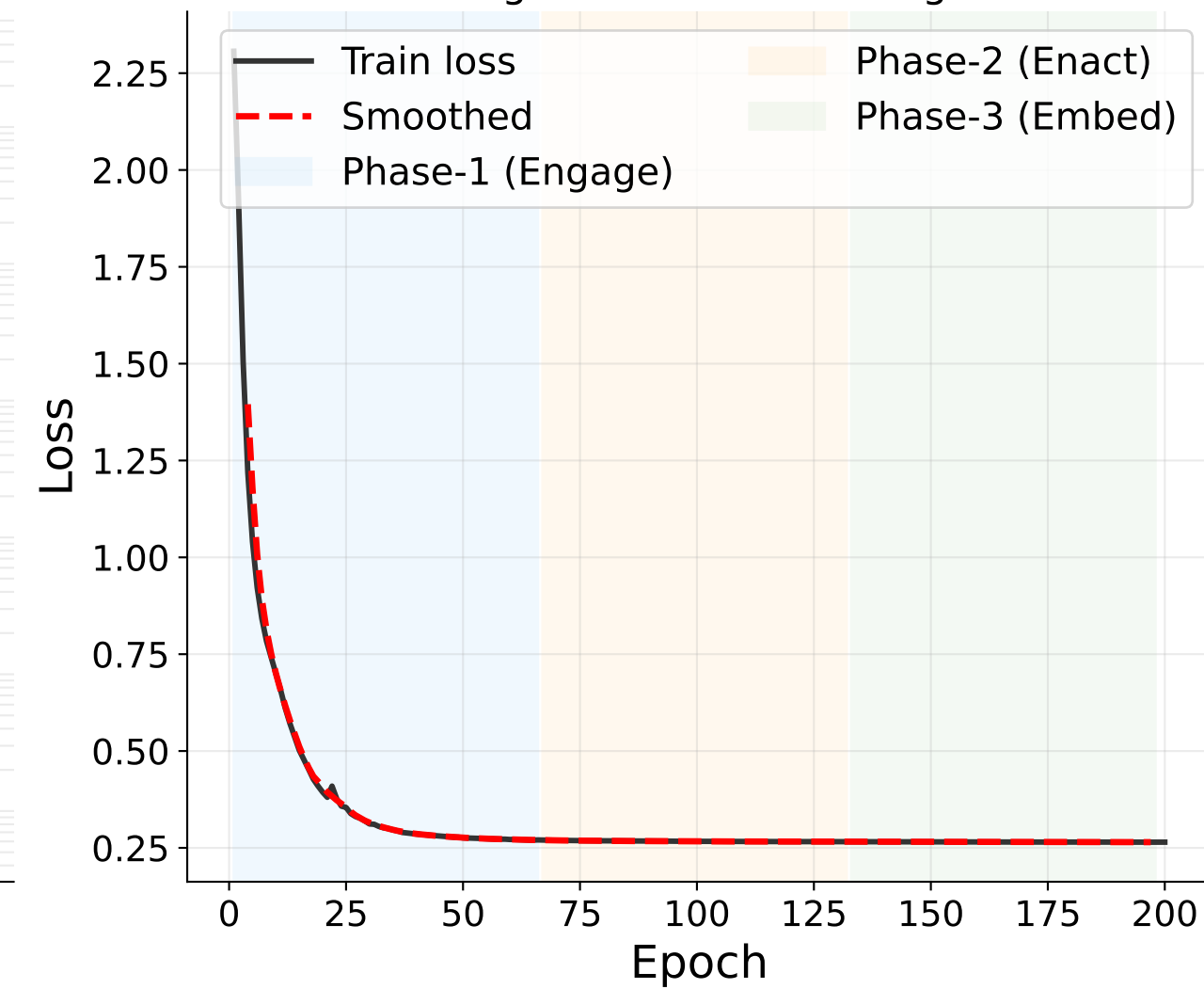
Gradient Norm per Phase
(Theorem 1: Engage bounds λe^{-L})



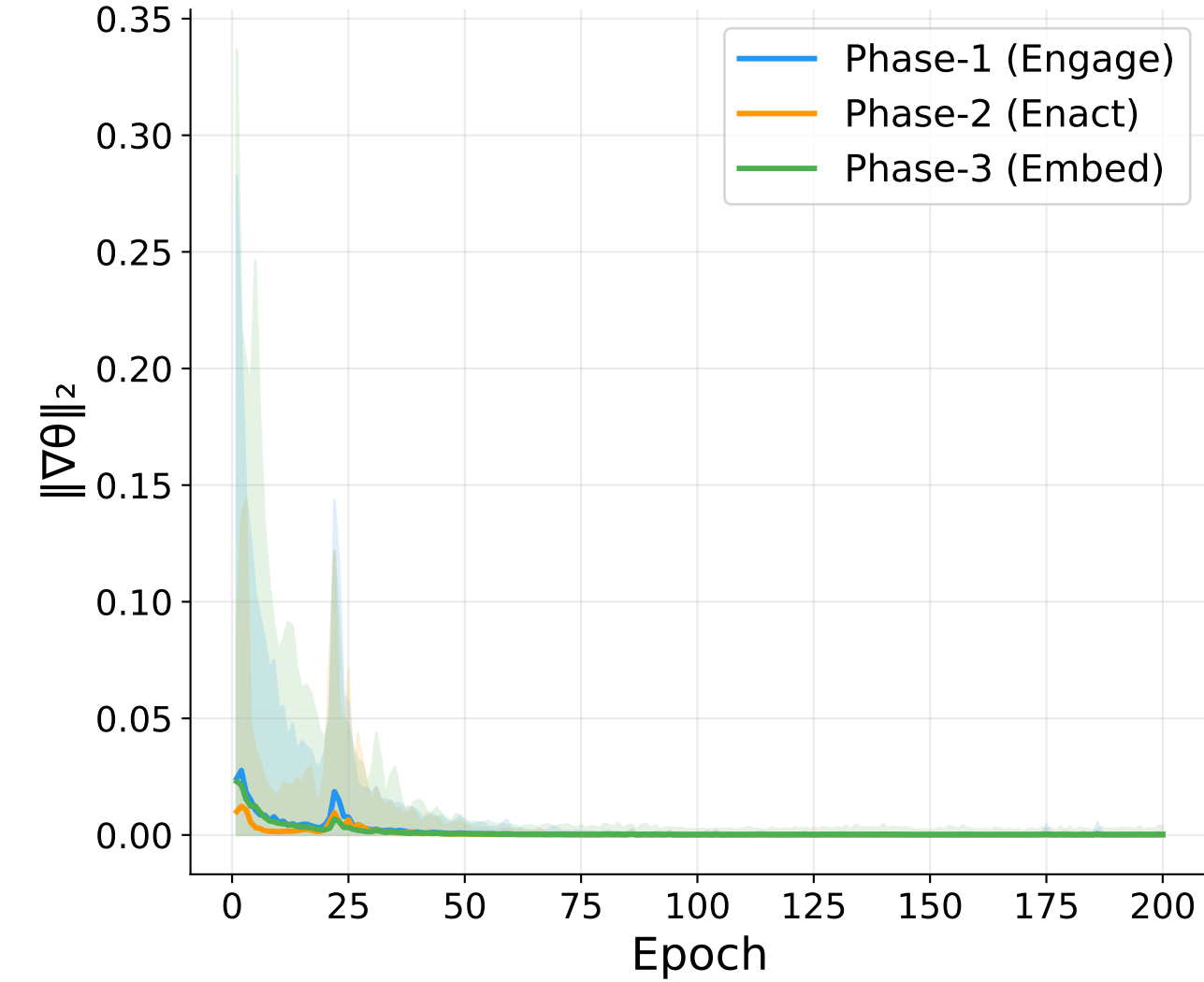
Gradient Variance per Phase
($\text{Var}[\nabla] \leq L^2/N$)



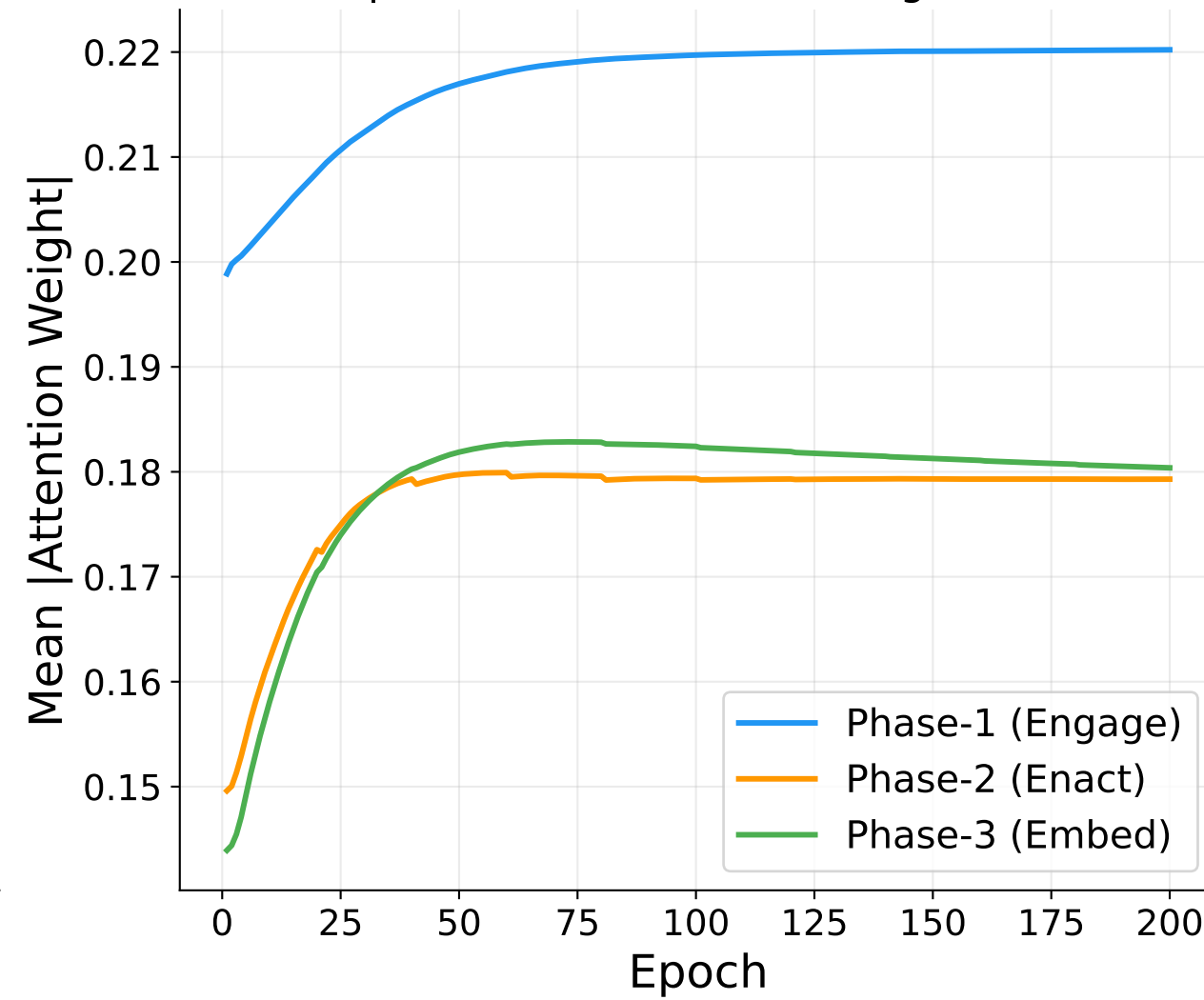
Training Loss with Phase Regions



Gradient Band (min / mean / max)



Attention / Gate Weight Magnitude
per Phase (s1/s2/s3, GAT, gates)



Relative Gradient Stability
($\div$ initial; 1.0 = stable)

